

difference between cs rs combined and separate,,,

Good question — this is **very important for understanding ADS8588H timing** 🔥

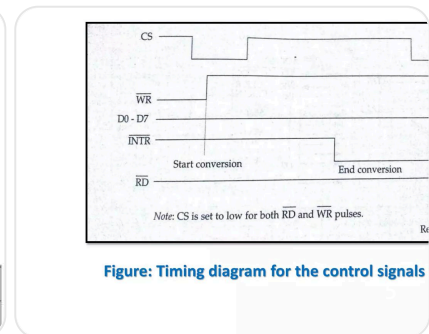
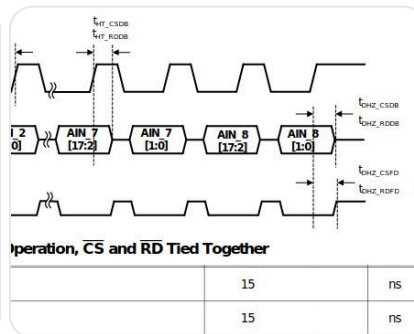
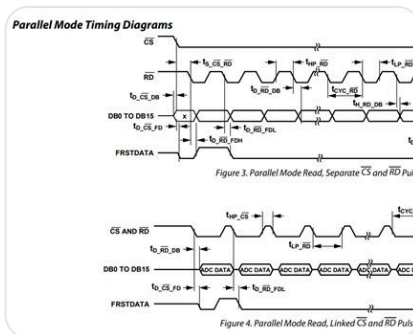
You're asking:

👉 CS & RD combined vs CS & RD separate — what is the difference?

🧠 FIRST: What are CS and RD?

- CS (Chip Select) → enables the ADC
- RD (Read) → tells ADC to output data

⚙️ CASE 1: CS & RD SEPARATE



🔧 How it works

- CS → LOW → ADC enabled
- RD → LOW → Data output

👉 You control them **independently**

🕒 Sequence

1. BUSY → LOW (data ready)
2. CS → LOW (select ADC)
3. RD → LOW → AIN1 data
4. RD pulse again → AIN2
5. Continue...

🎯 Key features

Key features

- ✓ Simpler hardware
 - ✓ Less control
 - ✗ No separation of enable & read
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Simple idea

👉 One signal does BOTH:

- Enable + Read
-

MAIN DIFFERENCE

Feature	Separate (CS & RD)	Combined (CS=RD)
Control signals	2	1
Flexibility	High	Low
Control over timing	Precise	Limited
Hardware complexity	More	Simple
Operation	Enable then Read	Both at same time

VERY IMPORTANT TIMING DIFFERENCE

Separate:

- You can do:
 - CS LOW first
 - wait
 - then RD LOW

👉 Safer timing control (respect tSU, tH)

Combined:

- CS & RD go LOW together

👉 Must satisfy:

- Setup timing automatically
 - Less margin for mistakes
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FINAL UNDERSTANDING

👉 Separate mode = professional, flexible control

👉 Combined mode = simple, quick control

Your ADC (ADS8588H) usage tip

- Use **separate CS & RD** if:
 - FPGA / precise timing
 - debugging timing issues
 - Use **combined** if:
 - simple MCU interface
 - less pins available
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If you want, next I can:

✓ Explain why Fig 2 vs Fig 4 show different timing behavior

✓ Or mark exact AIN1 → AIN2 transitions on your photo

Just tell 👍

actually cs chip select,, does it has any other chip? what actually means

Good — you're asking the **real meaning of CS (Chip Select)**, not just the signal behavior



Let's clear it **conceptually + practically**.

What does CS (Chip Select) actually mean?

👉 CS = "Hey, THIS chip... I'm talking to YOU now"

It is a **selection signal**, not a data signal.

Why do we need CS?

Because in real circuits:

👉 One microcontroller / FPGA is connected to **many devices**

Example:

- ADC (ADS8588H)
- DAC
- EEPROM
- Display
- Sensors

All may share:

- Data bus (DB[15:0])
 - Control lines
-

🔌 Without CS (Problem)

If no CS:

👉 All chips try to respond **at same time**

- ✗ Data collision
 - ✗ Wrong values
 - ✗ Bus damage
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✅ With CS (Solution)

- 👉 Only the selected chip responds
 - 👉 Others stay **silent (High-Z / disconnected)**
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🧩 Simple Analogy

Think classroom:

- Teacher = microcontroller
 - Students = chips
 - 👉 CS = calling a student's name
 - "Rahul!" → only Rahul answers
 - Others stay quiet
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⚡ In ADS8588H (Your ADC)

- 👉 CS LOW → ADC is active
 - 👉 CS HIGH → ADC ignores everything
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Important internal behavior

When CS = HIGH:

- Data bus → High impedance (Z)
 - ADC does NOT drive output
- 👉 This is **very important for shared bus systems**
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Your question:

“Does it have any other chip?”

- 👉 Not mandatory — but **designed for that possibility**
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Case 1: Single chip system

- Only ADS8588H connected
- 👉 Still CS is used:
- To **enable/disable output**
 - To control timing properly
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Case 2: Multiple chips (real-world use)

Example:

- 2 ADCs
- 1 DAC

Device	CS Line
ADC1	CS1
ADC2	CS2
DAC	CS3

- 👉 MCU selects one at a time
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 **VERY IMPORTANT POINT (Often missed)**

👉 CS is NOT just "selection"

👉 It also:

- ✓ Controls **when data appears**
 - ✓ Helps satisfy timing (tSU, tH)
 - ✓ Resets read sequence (in some modes)
-



In your timing diagrams (ADS8588H)

- CS LOW → enables output stage
- RD LOW → triggers data output

👉 If CS is HIGH:

✗ RD does nothing



FINAL UNDERSTANDING

👉 CS means:

- ✓ Select this chip
 - ✓ Allow it to communicate
 - ✓ Disconnect it when not needed
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One-line summary

👉 CS = Permission signal for a chip to talk on the bus